LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Vibration of rotor and gearbox vibrators, disc cause effects on helicopter monitoring devices and remedial action.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Vibration of rotor and gearbox vibrators, disc cause effects on helicopter monitoring devices and remedial action. On hand experience to use vibration measuring equipment's.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstrate use of vibration measuring equipment on simulated rotor/gearbox components; Safety: Wear PPE, ensure equipment is calibrated; Supervision: Instructor monitors probe placement and data reading.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary sources of vibration in a helicopter rotor system?
(b) How does excessive vibration affect helicopter monitoring devices?
(c) List three common causes of gearbox vibrator issues.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Main rotor movement dynamics • Translation flight vibration characteristics • Power transmission vibration effects • Auto rotation vibration behavior • Standard vibration check procedures	L & Ds	PPT, IB	15 min
(b)	• Disc causes of rotor vibration • Gearbox vibrator failure modes • Impact on monitoring device accuracy • Interpretation of vibration data trends • Remedial actions for identified faults	L & Ds	PPT, PDF	20 min
(c)	• Operation of vibration measuring equipment • Calibration and zeroing procedures • Probe placement on rotor and gearbox • Reading and recording vibration levels • Troubleshooting equipment malfunctions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between disc imbalance and monitoring device readings.
(b) Describe the remedial action required if gearbox vibration exceeds limits.

12. Home Assignment and Follow Up	Prepare a brief report on the vibration check procedures for the main rotor, referencing the next lesson plan (LP 02) on detailed rotor balancing adjustments.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Major, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min